SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I _BANAGANAPALLI MUNNA(192372028)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B Munna

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher
A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.
B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.
B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network
Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.
B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security
An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.
B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices
A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.
B. Recommend a suitable cryptographic approach and justify your answer.

1. Secure Data Transmission using Block Cipher

Scenario Understanding

A company is transmitting repeated financial data blocks over a network. They initially used Electronic Codebook (ECB) mode and observed repeated patterns in the ciphertext. Since financial information is confidential, the encryption method should prevent attackers from identifying patterns.

A. Explain why this issue occurs

Key Issue Identified

ECB mode encrypts each plaintext block independently using the same secret key.

Explanation

- Identical plaintext blocks produce identical ciphertext blocks.
- If the financial data contains repeated records (such as account numbers, transaction types, or repeated values), those blocks generate the same ciphertext.
- These repeated ciphertext blocks reveal patterns, allowing an attacker to infer information about the original data even without knowing the encryption key.
- Therefore, ECB does not provide semantic security for repeated data and is unsuitable for transmitting sensitive financial information.

Example

Plaintext Block	ECB Ciphertext
Payment	A1B2
Payment	A1B2
Salary	C3D4
Payment	A1B2

The repeated ciphertext (A1B2) clearly reveals repeated plaintext blocks.

Conclusion

The issue occurs because ECB encrypts identical plaintext blocks into identical ciphertext blocks, exposing data patterns and reducing confidentiality.

B. Suggest a more secure block cipher mode and justify your choice

Recommended Mode: Cipher Block Chaining (CBC)

Application of Concept

CBC mode uses an Initialization Vector (IV) and combines each plaintext block with the previous ciphertext block before encryption.

As a result:

- Identical plaintext blocks produce different ciphertext blocks.
- Ciphertext patterns are removed.
- It provides much stronger confidentiality than ECB.

Why CBC is Better

- Eliminates repeated ciphertext patterns.
- Protects sensitive financial information.
- Makes it much more difficult for attackers to analyze encrypted data.

Alternative

For modern systems, AES-GCM (Galois/Counter Mode) is even better because it provides:

- Confidentiality (encryption)
- Integrity (detects unauthorized changes)
- Authentication (verifies data authenticity)

Decision

For this scenario, CBC is a secure replacement for ECB, while AES-GCM is the preferred modern choice because it provides both encryption and authentication.

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5. Cryptography for IoT Devices

Scenario Understanding

A smart home system contains low-power IoT devices that must communicate securely. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices

Key Issues Identified

IoT devices have several resource limitations:

1. Limited processing power
   - Small processors cannot efficiently perform complex cryptographic calculations.
2. Limited memory
   - RAM and storage are very limited.
3. Low battery power
   - Devices often operate on batteries, requiring energy-efficient algorithms.
4. Limited bandwidth
   - Smaller keys and messages reduce communication overhead.
5. Real-time communication
   - Encryption and decryption should be completed quickly with minimal delay.

Interpretation

Any cryptographic algorithm chosen should provide strong security while minimizing computation, memory usage, power consumption, and communication overhead.

B. Recommend a suitable cryptographic approach and justify your answer

Recommended Approach: Elliptic Curve Cryptography (ECC)

Application of Concept

ECC provides the same security as RSA using much smaller key sizes.

For example:

- RSA → 1024-bit key
- ECC → 160-bit key (equivalent security)

Justification

1. Smaller keys
   - Require less storage and memory.
2. Lower computational cost
   - Faster cryptographic operations on low-power processors.
3. Lower power consumption
   - Extends battery life of IoT devices.
4. Lower communication overhead
   - Smaller keys and signatures reduce bandwidth usage.
5. Strong security
   - Maintains equivalent security with significantly fewer resources than RSA.

Decision

ECC is the best choice for the smart home IoT system because it offers:

- Equivalent security to RSA.
- Lower computation.
- Lower memory usage.
- Lower battery consumption.
- Better performance on resource-constrained devices.

Final Conclusion

1. Secure Data Transmission using Block Cipher

- Problem: ECB encrypts identical plaintext blocks into identical ciphertext blocks, exposing patterns.
- Recommended Solution: Replace ECB with CBC, or preferably AES-GCM, to hide patterns and provide stronger security.

2. Cryptography for IoT Devices

- Constraints: Limited CPU, memory, battery power, storage, and bandwidth.
- Recommended Cryptosystem: ECC, because it provides equivalent security to RSA while using much smaller keys, requiring less computation, memory, bandwidth, and power, making it ideal for low-power IoT devices.

RUBRICS FOR CALCULATION:

Criteria	Weighta ge (%)	Level 4 (Exempla ry)	Level 3 (Proficient)	Level 2 (Developi ng)	Level 1 (Begin nin g)
Understa ndi ng of Scenario	20%	Demonstr at es clear and complete understa ndi ng of the scenario and context	Shows good understan di ng with minor gaps	Partial understa ndi ng; misses key aspects	Misinter pre ts or fails to understa nd the scenario
Identificati on of Key Issues	20%	Accuratel y identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicati on of Concepts	25%	Applies appropria te concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand